

SMT. CHM COLLEGE | CAPSTONE VIVA VOCE EXECUTIVE DEFENSE CHEAT SHEET

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1. CORE MATHEMATICAL FORMULATIONS & VERIFIED ML TELEMETRY

Model / Protocol	Mathematical Formulation & Rule	Observed Benchmark Telemetry
Defaulter Classifier	$\sigma(z) = 1 / (1 + e^{-z}) \mid z = w^T x + b$	Accuracy: 98.3% Precision: 92.2% Recall: 96.6% F1: 0.943
Ordinance 0.119	Attendance $\geq 75.0\%$ (Mandatory Clear) $< 65.0\%$ (Debarment)	Real-time animated SVG gauge; Hall Ticket issuance gate
Anti-Cheating Seating	$D_{adj}(S_i, S_j) = 1 \blacksquare \text{Subject}(S_i) \neq \text{Subject}(S_j)$	0.0% adjacent collusion across Kundnani, Lab 301, Comm 204
Placement CTC Predictor	$CTC = 0.52(CGPA) + 0.045(Code) + 0.55(Certs) + 0.35(Intern) - 1.20$	Forecasts ■3.5 to ■14.0 LPA; Tier 1 Dream Offer matching
Anti-Proxy Dynamic QR	$Token(t) = HMAC\text{-}SHA256(LectureID \parallel Secret, \blacksquare t / 10 \blacksquare)$	10-second rotating ephemeral tokens eliminating screenshot proxy

2. INSTITUTIONAL RETURN ON INVESTMENT (ROI) & IMPACT NUMBERS

Annual Recurring Savings: ■ 18,50,000 / Year	Faculty Time Reclaimed: 4,500+ Instructional Hours / Year	Cloud Performance: 96+ Google Lighthouse < 400ms Initial Load
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3. TOP 5 RAPID-FIRE VIVA QUESTIONS & PREPARED VERBAL DEFENSES

Q1: Why vanilla web standards instead of React / Next.js?

Answer: Zero vendor lock-in, zero external framework runtime overhead, sub-400ms load time, 96+ Google Lighthouse score, and zero long-term maintenance cost for the institution.

Q2: How does the Dynamic QR token eliminate proxy attendance?

Answer: Classrooms project an HMAC-SHA256 token that rotates every 10 seconds. Static screenshots shared over messaging apps expire before remote students can scan them.

Q3: How does Module 23 AI Seating mathematically prevent cheating?

Answer: Bipartite checkerboard algorithm interleaves 4 programs (Data Science, IT, B.Com, BMS) ensuring $D_{adj} = 1$ has $\text{Subject}(i) \neq \text{Subject}(j)$, giving mathematically verified 0.0% adjacent collusion.

Q4: What happens during a campus network outage?

Answer: Progressive Web App service worker (sw.js) pre-caches student ID cards, timetables, and emergency contacts in chm-cache-v1, guaranteeing full offline functionality.

Q5: How is student privacy protected in your AI models?

Answer: 100% client-side execution in JavaScript. Zero student data leaves the device to external 3rd-party cloud APIs, satisfying statutory data privacy standards.